

Comparing Models, Qualifying Conclusions

Answer Key

Precalculus / Statistics

Quick Answers

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| 1. 8 mm | 9. (a) the sum of squared residuals for Model B = 5.5, — |
| 2. 8 mm | 10. (a) bus, given grade 9 = $\frac{2}{5}$, (b) bus, given grade 10 = $\frac{1}{4}$, — |
| 3. -0.6 cm | 11. (a) Model A at week 8 = 27.4 cm, (b) Model B at week 8 = 26.5 cm |
| 4. (a) the proportion who walk = $\frac{13}{40}$, — | 12. (a) the gap between the two predictions = 0.9 cm, — |
| 5. (a) the mean = 10 mm, — | |
| 6. 1.39 | |
| 7. — | |
| 8. (a) households surveyed = 140, (b) the modal number of cars = 1 | |
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What is verified

14 of 20 answers machine-checked against SymPy, across 12 problems.

— marks an answer only you can judge — problems 4, 5, 7, 9, 10, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus / Statistics

HSS-IC.A – *problems 4, 10*

HSS-IC.B – *problems 7, 12*

HSS-ID.A – *problems 1, 2, 5, 8*

HSS-ID.B – *problems 3, 6, 9, 11*

Difficulty 1–5 of 5 across 20 tagged checks.

- 1.** Data set 1 (2, 4, 6, 7, 8, 10, 12, 14, 27): the median.

Solution: The nine values are already in order, so the median is the fifth: four below it and four above. That value is 8.

8 mm

- 2.** Data set 1: the interquartile range.

Solution: Split at the median and leave it out of both halves. Lower half 2, 4, 6, 7 has median $Q_1 = \frac{4+6}{2} = 5$. Upper half 10, 12, 14, 27 has median $Q_3 = \frac{12+14}{2} = 13$. So $\text{IQR} = 13 - 5$.

8 mm

The IQR ignores the 27 entirely — that is exactly why it pairs with the median rather than with the mean.

3. Data set 2: the residual of Model A at week 2.

Solution: Model A predicts $\hat{y} = 1.0 + 3.3(2) = 1.0 + 6.6 = 7.6$ cm, and the plant actually measured 7 cm. Residual is observed minus predicted: $7 - 7.6$. -0.6 cm

A negative residual means the model over-predicted — the line sits above the point at that week.

4. Travel survey ($n = 200$). (a) Proportion who walk. (b) Which population does it describe?

Solution (a): The Walk column totals $30 + 35 = 65$ out of 200 students, and $\frac{65}{200}$ reduces by 5 to $\frac{13}{40}$. $\frac{13}{40}$

Solution (b) — instructor-judged, no machine check. The survey covered grades 9 and 10 at one school, so $\frac{13}{40}$ describes those 200 students directly, and — if they were a random sample of those two grades — can be extended with caution to all grade 9 and 10 students at that school. It must not be extended to the whole school, since grades 11 and 12 were never asked, nor to students at other schools. Full credit names the population the figure does describe and at least one group it does not. Do not accept “the sample is too small”: 200 is a decent sample, and coverage, not size, is the problem here.

5. Data set 1. (a) The mean. (b) Which summary describes a typical day?

Solution (a): The nine values sum to $2 + 4 + 6 + 7 + 8 + 10 + 12 + 14 + 27 = 90$, and $90 \div 9 = 10$. 10 mm

Solution (b) — instructor-judged, no machine check. The mean is 10 mm and the median is 8 mm. The gap is caused by the single 27 mm day: it enters the mean at full weight and drags it up, while the median only cares about position and barely moves. For a *typical* day the median is the better summary, and the spread quoted alongside it should be the IQR rather than the standard deviation. Full credit identifies 27 as the cause and chooses a summary with a reason. A defence of the mean is acceptable only if it acknowledges the outlier and explains why the total rainfall still matters.

6. Model A’s sum of squared residuals, from $-0.3, -0.6, 0.1, -0.2, 0.5, -0.8$.

Solution: Squaring removes the signs, so over- and under-predictions both count against the model:

$$0.09 + 0.36 + 0.01 + 0.04 + 0.25 + 0.64 = 1.39.$$

So Model A’s fit score is

1.39

7. Screen time and reading score: why the headline is not supported.

Solution — instructor-judged, no machine check. The study is observational: nobody assigned students their screen time, so the two variables were free to move together for reasons that have nothing to do with one causing the other. A strong negative correlation across 500 students is real evidence that the two go together, and no evidence at all about direction. The response must say that observational data cannot establish causation *and* name one plausible confounder — hours of sleep, family income, or simply how much free time a student has, each of which could push screen time up and reading scores down independently. Reverse causation is also open: weaker readers may choose screens over books. Do not accept a bare “correlation is not causation” with no variable named and no reason.

8. Chart A. (a) Households surveyed. (b) The modal number of cars.

Solution (a): Add the four bars: $18 + 62 + 45 + 15$.

140

Solution (b): The tallest bar is the one at 62 households, which is the category for one car.

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The mode is a third summary model for “typical”, alongside the mean and the median in problems 1 and 5 — and for a count like this it is the only one of the three that is guaranteed to be a value the data actually takes.

9. Model B’s residuals $-1.5, -1.5, -0.5, -0.5, 0.5, -0.5$. (a) Its fit score. (b) Which model wins, and what that does not license.

Solution (a):

$$2.25 + 2.25 + 0.25 + 0.25 + 0.25 + 0.25 = 5.5.$$

So Model B’s fit score is

5.5

Solution (b) — instructor-judged, no machine check. Model A wins: 1.39 against 5.5, so it sits closer to the six measured weeks. The qualification that must appear is that fitting weeks 1–6 better says nothing reliable about week 20. That is extrapolation far outside the range of the data, and a sunflower’s height cannot keep rising by 3.3 cm a week forever — the straight line is a description of a window, not a law. Full credit picks Model A with the two numbers as evidence and states the extrapolation caveat; award partial credit for the choice alone.

10. Travel survey. (a) Bus among grade 9. (b) Bus among grade 10. (c) What the comparison supports.

Solution (a): Restrict to the grade 9 row: $\frac{40}{100}$, which reduces to

$\frac{2}{5}$

Solution (b): The grade 10 row gives $\frac{25}{100}$, which reduces to

$\frac{1}{4}$

Solution (c) — instructor-judged, no machine check. Within this survey a larger share of grade 9 students took the bus, $\frac{2}{5}$ against $\frac{1}{4}$. The response must stop there. Each group has 100 students and nothing in the data says how much a gap of this size varies from one survey to the next, so the comparison describes these respondents rather than establishing that grade 9 students in general prefer the bus. Full credit reports the comparison in the right direction *and* declines to generalise, with a reason. Do not accept a flat “grade 9 students prefer the bus”.

11. Week-8 predictions. (a) Model A. (b) Model B.

Solution (a): $\hat{y} = 1.0 + 3.3(8) = 1.0 + 26.4$.

27.4 cm

Solution (b): $\hat{y} = 2.5 + 3.0(8) = 2.5 + 24$.

26.5 cm

Inside the data the two lines are hard to tell apart; two weeks past the last measurement they have already separated.

12. Synthesis. (a) The gap between the two week-8 predictions. (b) A defensible two-sentence conclusion.

Solution (a): $27.4 - 26.5$, or directly $(1.0 + 3.3(8)) - (2.5 + 3.0(8))$.

0.9 cm

The gap grows by 0.3 cm for every further week, because the two slopes differ by 0.3.

Solution (b) — instructor-judged, no machine check. A defensible answer says something the data supports — over weeks 1 to 6 the sunflower grew at roughly 3 cm per week, and a straight line describes those six measurements closely, Model A most closely — followed by a genuine qualification. Any one of these counts: the models were fitted to six measurements of a single plant, so they say nothing about sunflowers in general; week 8 is already outside the observed range and the two models disagree there by 0.9 cm, a gap that widens by 0.3 cm every further week; or growth cannot stay linear indefinitely. Full credit is a supported claim plus one real qualification, in two sentences. An unqualified prediction for week 20 earns nothing — that is the exact move the whole sheet is built to prevent.